Department of Electrical & Electronic Engineering
Brac University
Semester: Spring 25

EEE101L

ELECTRICAL CIRCUITS I LABORATORY

Section:05

Software Experiment 08

Prepared by: Alif Tamjid & Souvik Barman Ratul

Group Number: 05

Group members:

SL	Student ID	Name
01.	24104215	Shreya Das
02.	24121205	Souvik Barman Ratul
03.	24121280	Tanisha Hannan
04.	24121308	Alif Tamjid

Objective: Maximum power transfer Theorem and RC Circuit Simulation in PSpice circuit.

Part-A: Maximum power transfer Theorem

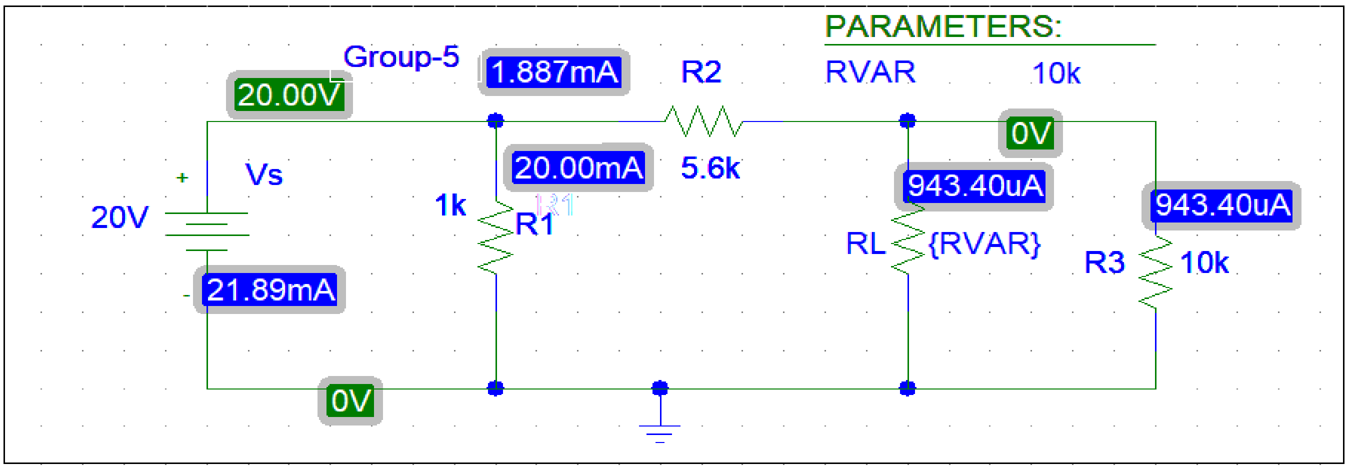


Fig-1

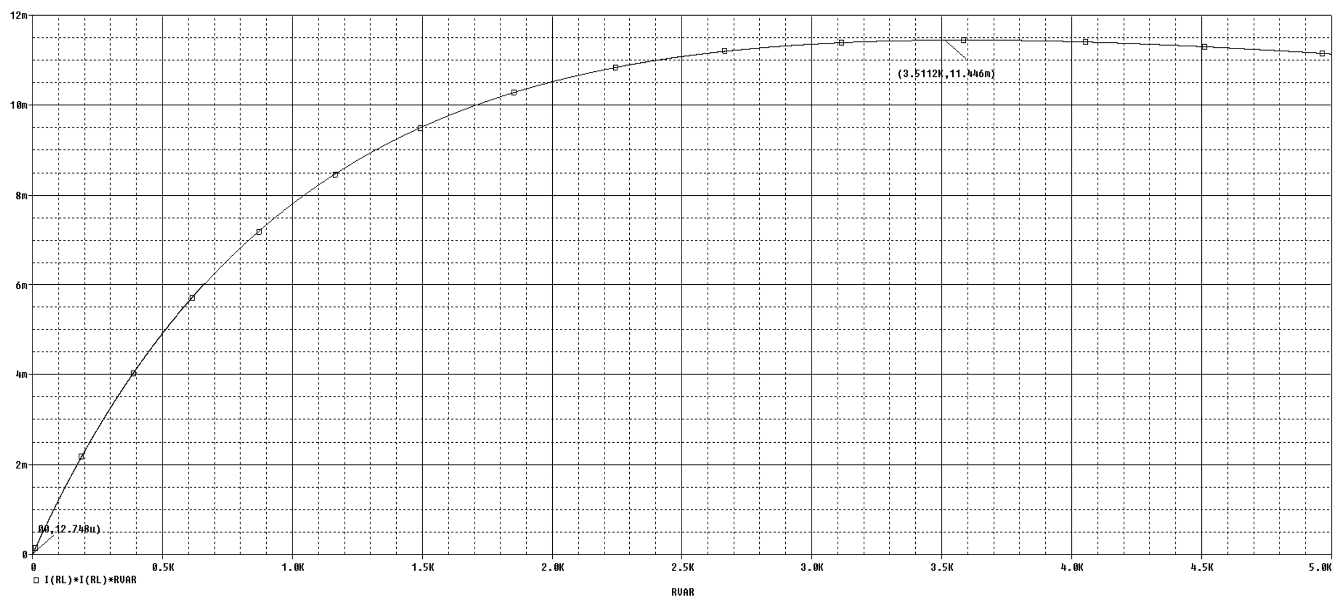


Fig-2

Here,

Rth= 3.5k

Part-B: RC Circuit Simulation

Given that,
f = 250 Hz
Vsource=6V (peak to peak)
DC OFFSET = 3V

Now,
V1=(3+3)=9V
V2=(-3+3)=0V
TD=0
TR=0.0001u
TF=0.0001u
PW=PER/2=0.004/2=0.002s
PER=1/f=1/250=0.004s

Time period for one cycle= 0.004s
Time period for four cycle=(0.004*4)= 0.016s=16ms

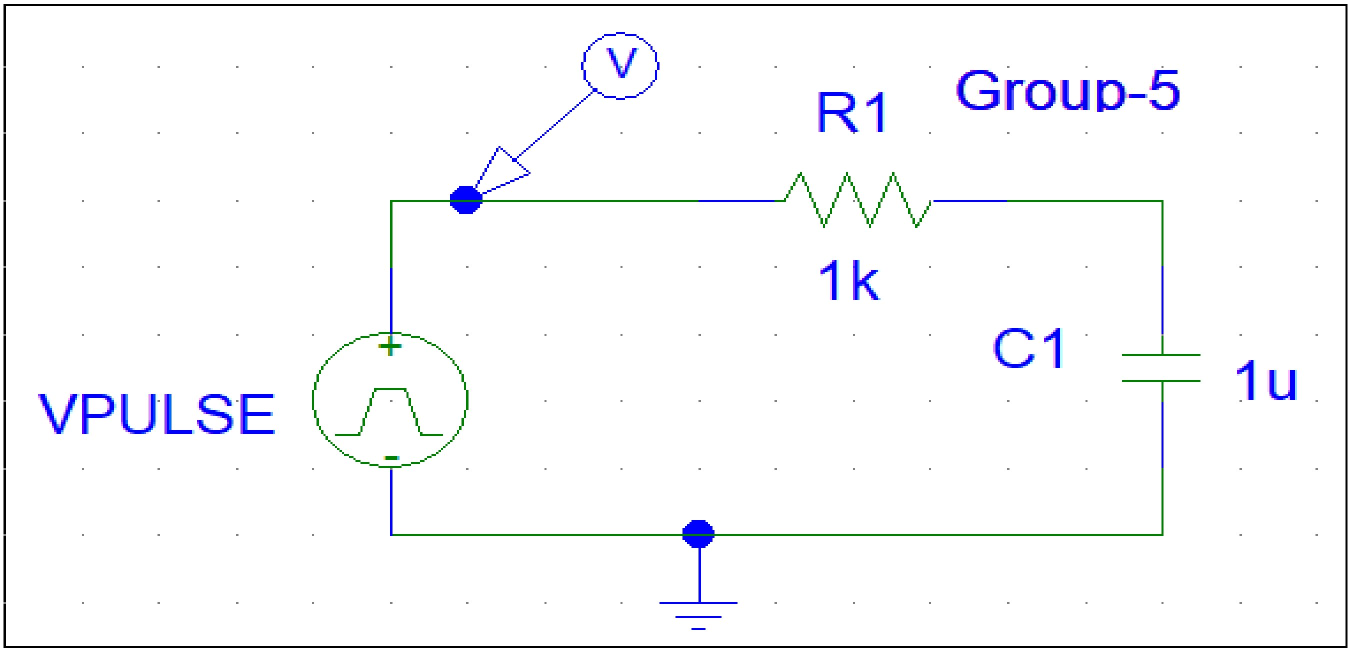


Fig-1

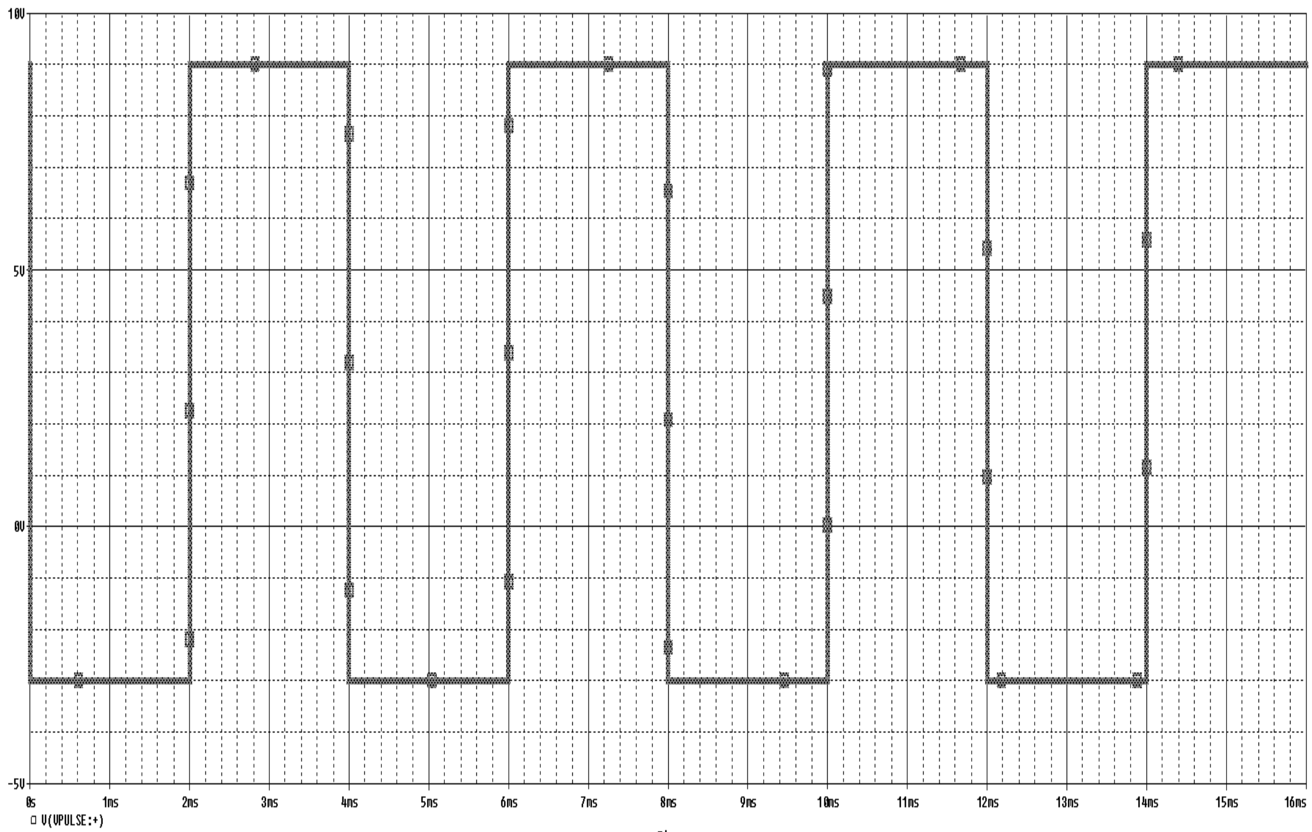
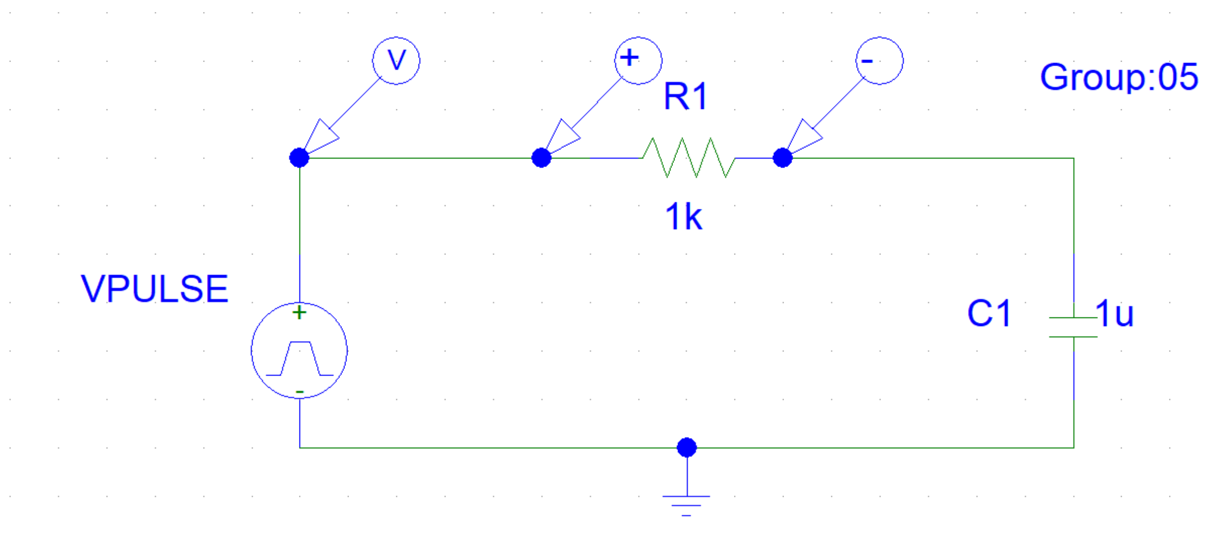
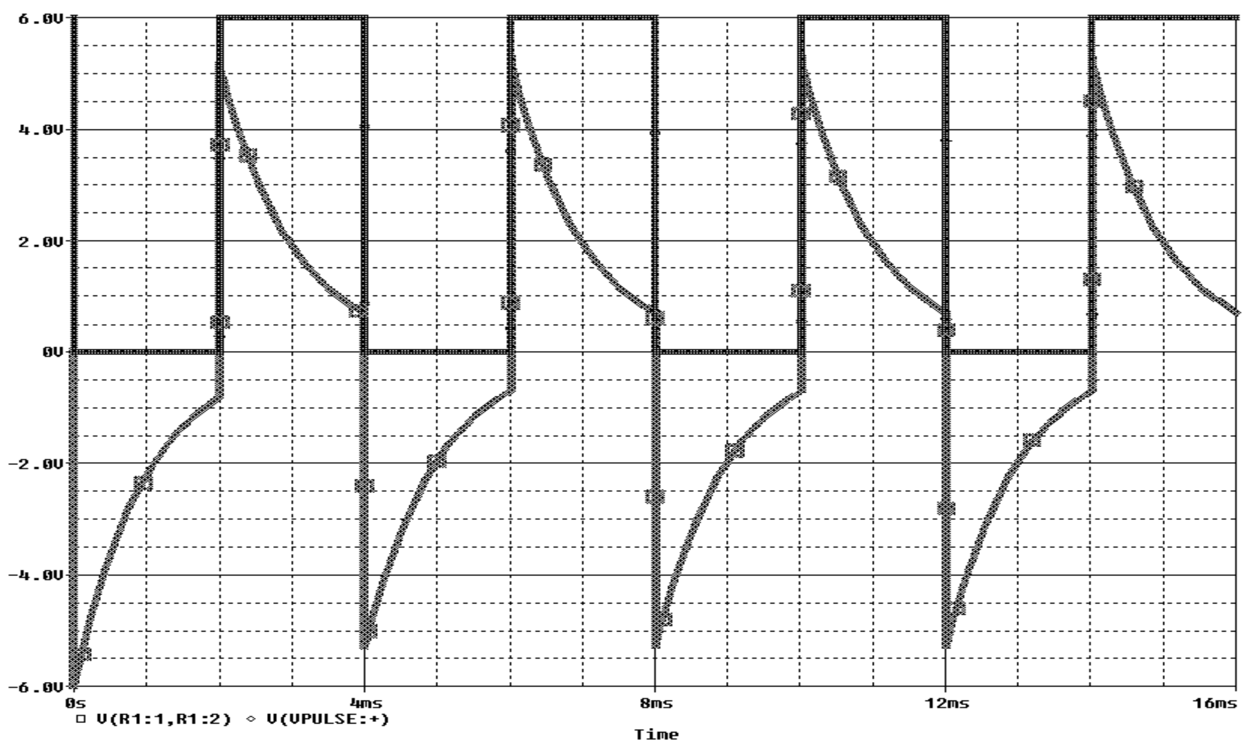


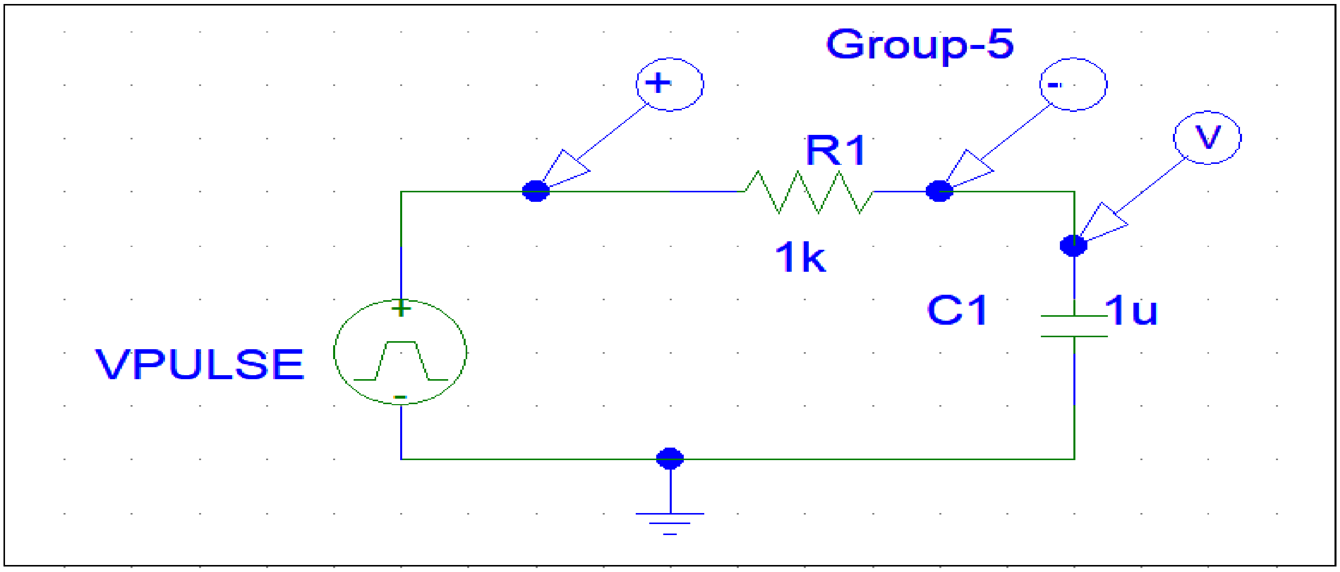
Fig-2 (Graph of Voltage source)

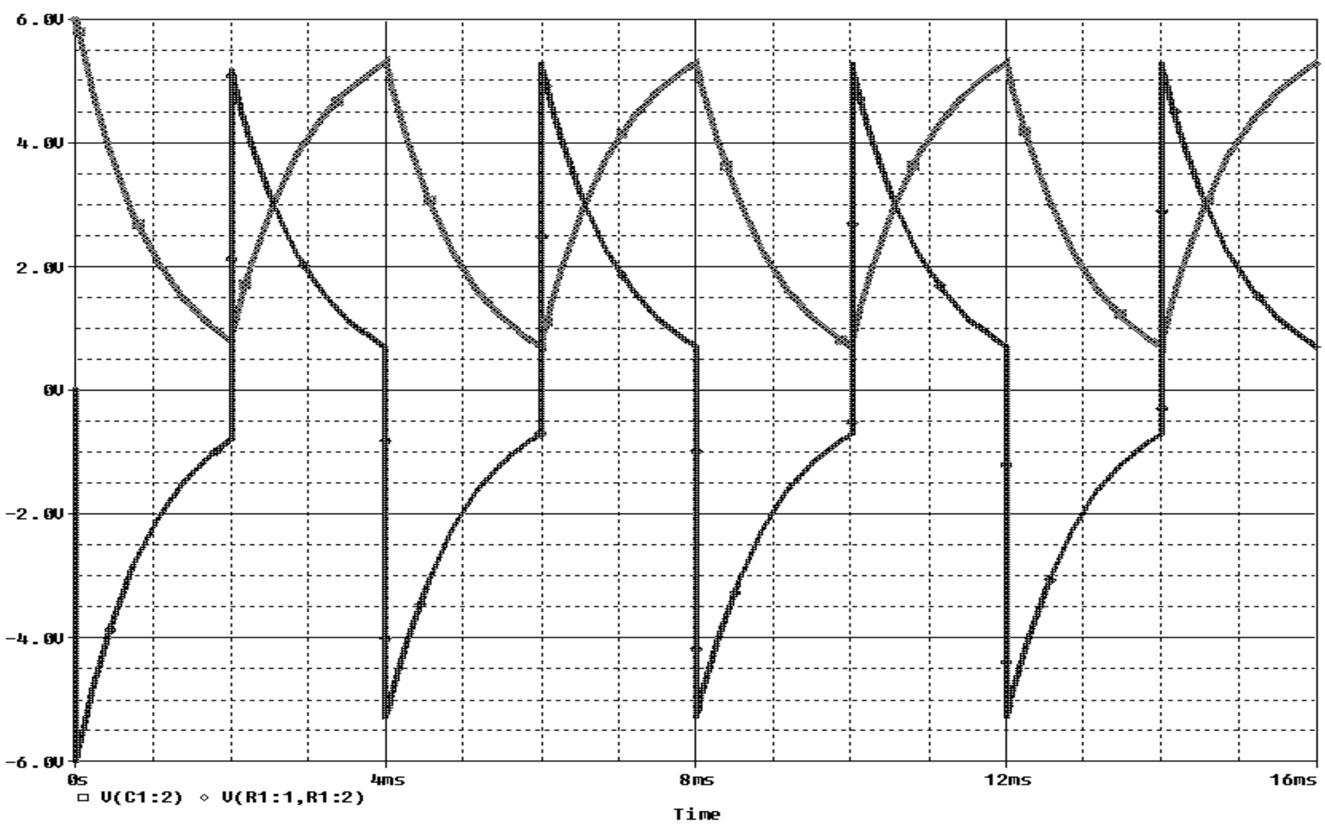
To-do list



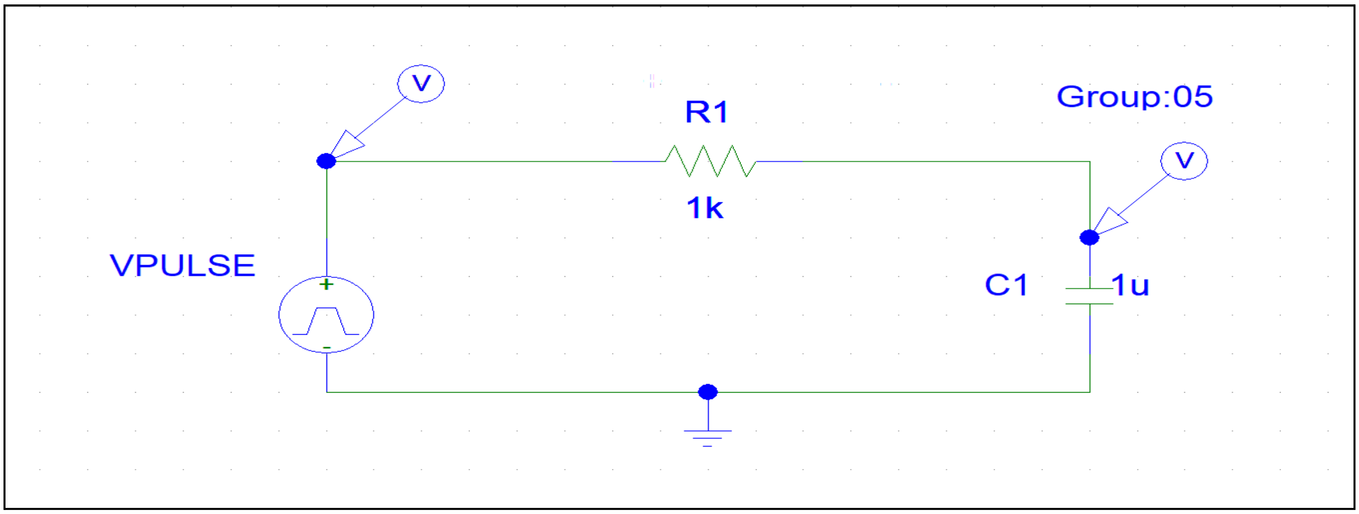


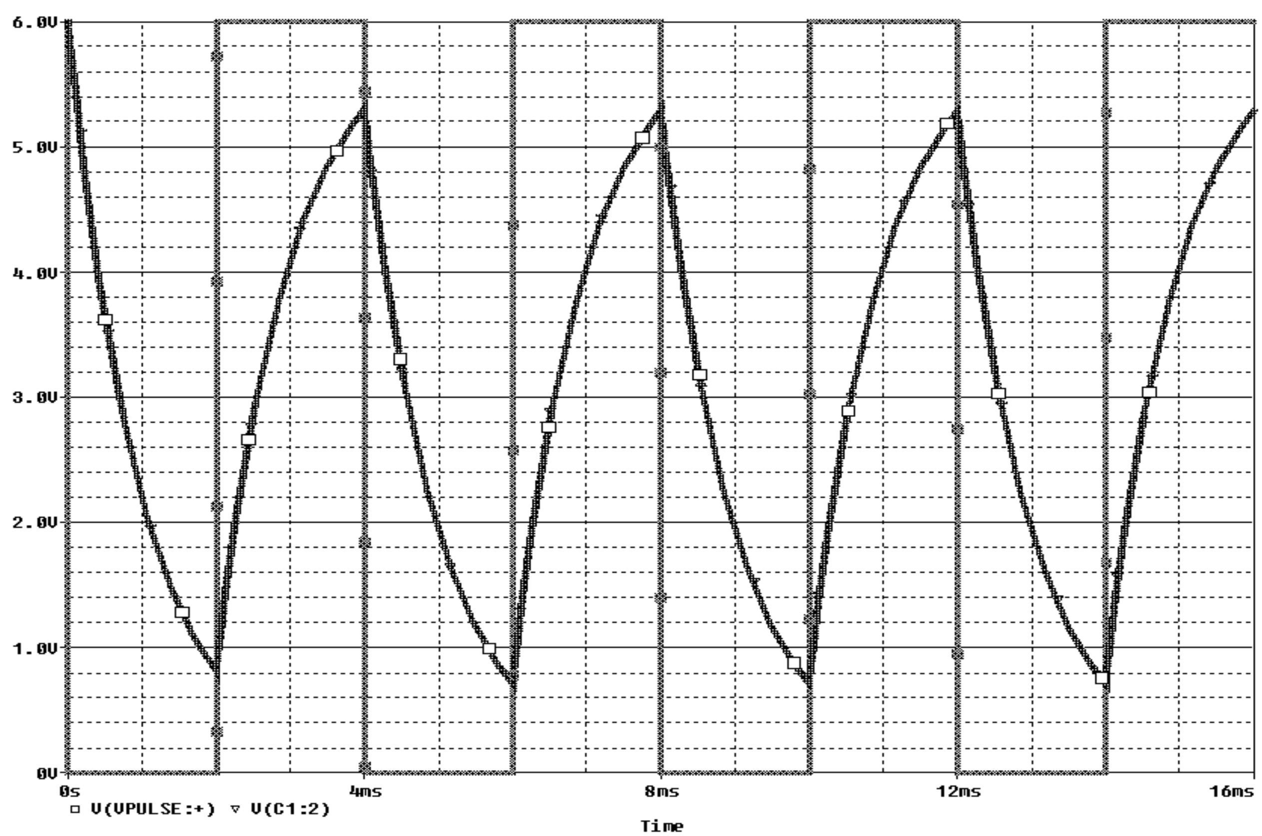
Graph of (Vs-VR)



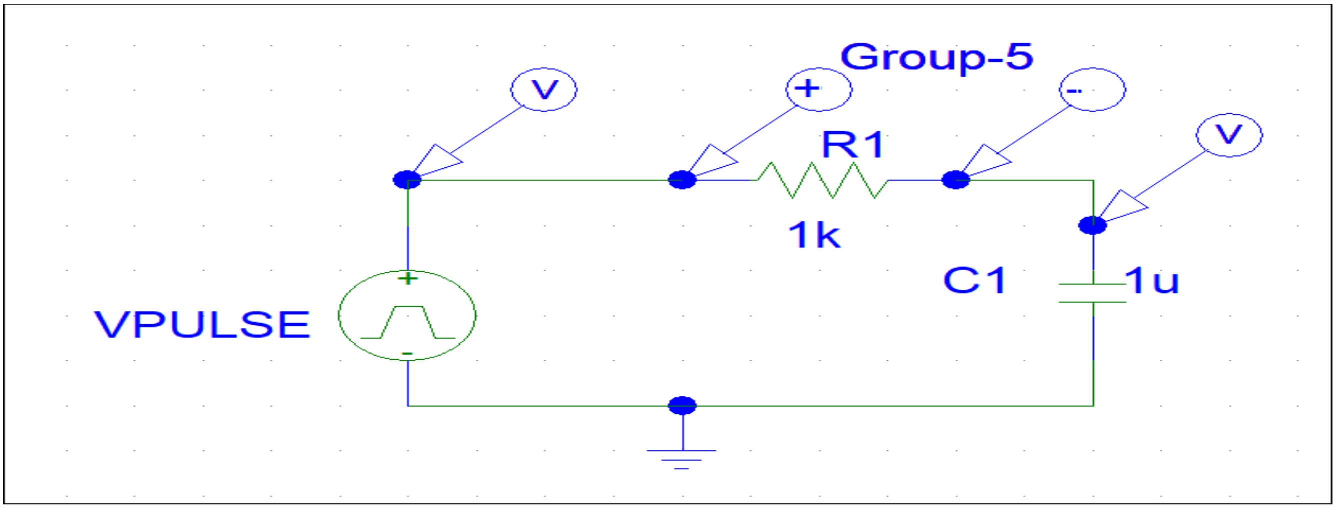


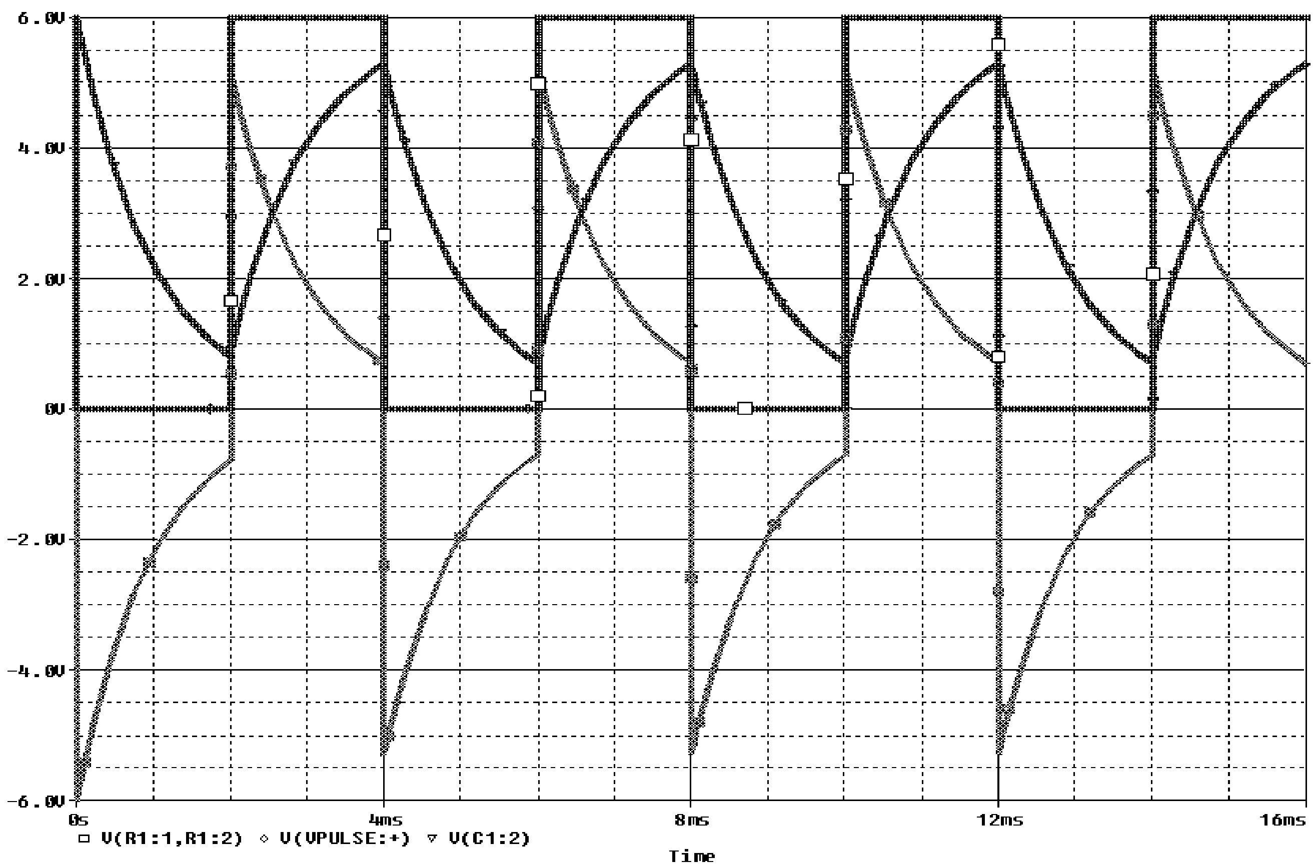
Graph of (V_R-V_c)





Graph of (Vs-Vc)





Graph of (V_s-V_R-V_c)

Conclusion:

In this experiment, we successfully explored the Maximum Power Transfer Theorem and performed RC circuit simulations using PSpice. Through the first part of the experiment, we verified that maximum power is transferred to the load when the load resistance equals the Thevenin resistance of the circuit. In the second part, we analyzed the transient response of an RC circuit under a square wave input, observing the charging and discharging behavior of the capacitor. The graphs generated from the simulation: such as (Vs - Vr), (Vr - Vc), (Vs - Vc) and (Vs-Vr-Vc) which provided valuable insights into the voltage distribution across each component. Overall, this experiment enhanced our understanding of circuit behavior in both theoretical and practical contexts, reinforcing core concepts of electrical engineering.